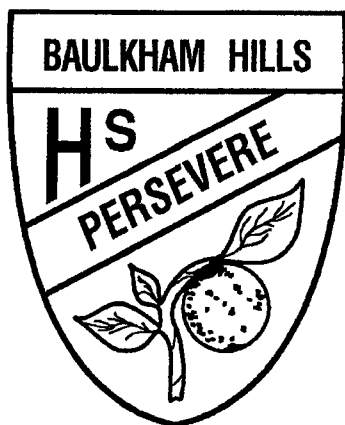




**Year 12
HSC Assessment Task
June 2017**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using non erasable black or blue pen
- Board-approved calculators may be used
- Show all necessary working
- Marks may be deducted for careless or badly arranged work
- Reference sheet is provided at the back of this paper.

Total marks – 48

Questions 1-3 (pages 2-4)

Answer each question on the appropriate pages of your answer booklet.

Question 1 (17 marks) - Use the Question 1 section of the writing booklet.		Marks
a)	Find $\int \frac{\sin x}{1+\cos x} dx$ using the substitution $u = \cos x$.	3
b)	Find $\int \frac{dx}{x(1+\ln x)^5}$ using the substitution $u = 1 + \ln x$.	3
c)	Find the exact value of $\int_{-1}^2 \frac{dx}{x^2+2x+10}$ using the substitution $u = x + 1$	3
d)	Given that $x = 0.7$ is an approximate root of the equation $\cos 2x = x$, use one application of Newton's method to obtain another approximation to this root. Give your answer correct to two decimal places.	3
e)	Given $f(x) = x - 4 + e^{2x}$. (i) Show that there is a zero between $x = 0.6$ and $x = 0.7$. (ii) Use one application of halving the interval method to find a smaller interval containing the zero. (iii) Hence solve $x = 4 - e^{2x}$ correct to one decimal place, given that $x = 4 - e^{2x}$ has only one solution.	2 2 1
End of Question 1		

Question 2 (16 marks) - Use the Question 2 section of the writing booklet.	Marks
a) A particle is moving so that its velocity at certain position x metres is given by $v^2 = 400 - 2x$. Particle is initially at the origin and travelling with a velocity of 20 metres per second. (i) Find x in terms of time t. (ii) Find when the particle is stationary. (iii) Find the time when the particle returns back to the origin. (iv) Describe the motion after the particle returns back to the origin	3 2 2 1
b) A steady wind is blowing with speed of 10 metres per second. From clouds moving horizontally with the wind, heavy raindrops fall to the ground 200 metres below. Air resistance is neglected and the approximate value of gravity g is 10 m/s^2. The diagram illustrates a physics problem involving projectile motion. A cloud is shown moving horizontally to the right with a constant velocity $v = 10 \text{ m/s}$. A raindrop falls from the cloud, following a parabolic path due to gravity. The vertical distance from the cloud to the ground is labeled as 200 m. The raindrop hits the ground with a splash, indicating the end of its fall. (i) Derive the equation for the horizontal (x) and vertical (y) displacement of a raindrop. (ii) Find the time when the drop hits the ground. (iii) Find the speed and acute angle at which the raindrop hits the ground.	3 2 3
End of Question 2	

Question 3 (15marks) - Use the Question 3 section of the writing booklet.		Marks
a)	A particle moves along the x-axis such that at time t seconds, the acceleration $\ddot{x}$ is given by $\ddot{x} = \frac{-1}{4(x-2)^3}$, where x is the displacement in metres from the origin. Find the expression of velocity v in terms of x, if initially the particle is 3 metres to the right from the origin travelling with the velocity $v = 0.5\text{m/s}$.	3
b)	A particle is moving in Simple Harmonic Motion, so that its displacement x centimetres from the origin at time t seconds is given by $x = 1 + 4 \cos(3t) + 4\sin(3t)$. (i) Express $\ddot{x}$ in the form $\ddot{x} = -n^2(x - x_0)$ where x_0 is the centre of the motion. 2 (ii) Find the period and amplitude of the motion. Answer in exact form. 2 (iii) Find the maximum distance of the particle from the origin. 1 (iv) Find the maximum acceleration of the particle. 1 (v) Find the speed of the particle when $x = 2$ cm. 2	
c)	Michelle is standing at a point M watching a balloon being released from a point W which is 100 metres away on the horizontal ground. The balloon rises vertically at a constant velocity of 5 metres per second. Let θ radians be the angle of elevation of the balloon at time t seconds and x metres be the distance the balloon has travelled in that time. (i) Find the expression for the rate of change of the angle of elevation of the balloon over time. Answer in terms of x. 3 (ii) Find the rate of change of the angle of elevation over time of the balloon when $\theta = \frac{\pi}{4}$. 1	
End of the Exam		

Question 1

		Marks
a) $\int \frac{\sin x}{1+\cos x} dx$	$u = \cos x$ $du = -\sin x dx$	3 - correct solns. 2 - obtains correct integrand in terms of u-only 1 - correct subst.
$= \int \frac{-du}{1+u}$		1 - finding du correctly 1 - correctly integrates
$= -\ln 1+u + C = -\ln 1+\cos x + C$		
b) $\int \frac{dx}{x(1+\ln x)^5}$	$u = 1 + \ln x$ $du = \frac{1}{x} dx$	3 - correct solns. 2 - obtains correct integrand in terms of u-only 1 - correct subst.
$= \int \frac{du}{u^5}$		1 - finding du correctly 1 - integrates correctly
$= \frac{u^{-4}}{-4} + C = -\frac{1}{4(1+\ln x)^4} + C$		
c) $\int_{-1}^2 \frac{dx}{x^2+2x+10}$	$u = x+1$ $u^2 = x^2+2x+1$ $du = dx$ $x=-1 \therefore u=0$ $x=2 \therefore u=3$	3 - correct solns. 2 - correct integrand in terms of u only with changed limits 1 - correct integrand in terms of u without changed limits 1 - integrates correctly
$= \int_0^3 \frac{du}{u^2+9}$		
$= \frac{1}{3} \left[\tan^{-1} \frac{u}{3} \right]_0^3$		
$= \frac{1}{3} \left[\tan^{-1} 1 - \tan^{-1} 0 \right] = \frac{1}{3} \times \frac{\pi}{4} = \frac{\pi}{12}$		

d) $\cos 2x = x \therefore$ $\cos 2x - x = 0$ let $f(x) = \cos 2x - x$	3 - correct solns.
$f'(x) = -2\sin 2x - 1$ if $x_0 = 0.7$	2 - finds a relevant function, its derivative & applies Newton's method correctly
$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$	1 - finds a relevant function
$x_1 = 0.7 - \frac{\cos(2 \times 0.7) - 0.7}{-2\sin(2 \times 0.7) - 1}$	1 - derives $f(x)$ correctly 1 - applies Newton's method correctly
$x_1 = 0.52159...$ $\therefore x_1 = 0.52$ (2 d.p.)	
e) i) $f(0.6) = -0.0798... < 0$ $f(0.7) = 0.7551... > 0$ since $f(0.6) < 0$ and $f(0.7) > 0$ and $f(x)$ is continuous $\therefore 0.6 < x < 0.7$ [x = zero]	2 - correct solns. 1 - finds $f(0.6)$ and $f(0.7)$ correctly
ii) $f\left(\frac{0.6+0.7}{2}\right) = f(0.65) = 0.31929...$ $\therefore 0.6 < x < 0.65$	2 - correct solns. 1 - applies the method correctly
iii) $f(x) = 0 \therefore 0 = x - 4 + e^{2x}$ $x = 4 - e^{2x}$	1 - correct solns.
from (ii) $0.6 < x < 0.65$ $\therefore x = 0.6$ (1 d.p.)	0 - no mark for bold answer of $x = 0.6$.

Question 2

Marks

a) Method ①	
i) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} (200 - x)$	3 - correct solus.
$\therefore \ddot{x} = -1 \text{ (m/s}^2\text{)}$	2 - derives $\ddot{x} = -1$
now $\dot{x} = \int \ddot{x} dt = \int -1 dt = -t + C$	and uses $\dot{x} = \int \ddot{x} dt$
$t=0 \quad \dot{x} = +20 \quad \left. \begin{array}{l} 20 = 0 + C \therefore C = 20 \end{array} \right\}$	to find v in terms of t
$\therefore \dot{x} = -t + 20$	
$x = \int \dot{x} dt = \int -t + 20 dt$	2 - for relevant method to find $\frac{dx}{dt}$
$x = -\frac{t^2}{2} + 20t + C$	
$t=0 \quad x=0 \therefore C=0 \quad \boxed{x = -\frac{t^2}{2} + 20t}$	2 - obtains $t = f(x)$ 1 - finds $\ddot{x} = -1$
or relevant Method ②	1 - applies rules
$v^2 = 400 - 2x$	$v = \frac{dx}{dt}, \frac{dt}{dx} = \frac{1}{v}$
$\therefore v = \pm \sqrt{400 - 2x}$	$t=0 \quad \ddot{x} = -1 \quad v=0$ $x=0 \quad v=+20 \quad x=200$
$\therefore v = \sqrt{400 - 2x}$ ①	1 - attempts to find t in terms of x
for $0 \leq x \leq 200$ $\boxed{200 \geq x}$	
$\therefore \frac{dx}{dt} = \sqrt{400 - 2x} = (400 - 2x)^{\frac{1}{2}}$	1 - significant progress towards solution
$\therefore \frac{dt}{dx} = (400 - 2x)^{-\frac{1}{2}}$	
$t = \int (400 - 2x)^{-\frac{1}{2}} dx$	

$\therefore t = \frac{(400 - 2x)^{\frac{1}{2}}}{\frac{1}{2} \times -2} + C$	$t=0 \quad x=0$ $\boxed{C=20}$
$\therefore t = 20 - (400 - 2x)^{\frac{1}{2}}$ ①	
$(400 - 2x)^{\frac{1}{2}} = 20 - t$	
$400 - 2x = (20 - t)^2$	
$400 - (20 - t)^2 = 2x$ ①	
$\boxed{200 - \frac{1}{2}(20 - t)^2 = x}$	or $\boxed{x = -\frac{t^2}{2} + 20t}$
ii) $v=0$	Marks
By using Method ① $\dot{x} = v = -t + 20$	2 - correct solus.
$t = 20$	
By using Method ② $v^2 = 400 - 2x$	1 - finds x when $v=0$
$0 = 400 - 2x$	
$\boxed{x = 200}$ ①	
sub. $x = 200 \therefore 200 = -\frac{t^2}{2} + 20t$	
$t = 20$	
or sub. in $t = 20 - (400 - 2x)^{\frac{1}{2}}$	
$\therefore \boxed{t=20}$ ①	

Marks

iii) $x = 0$

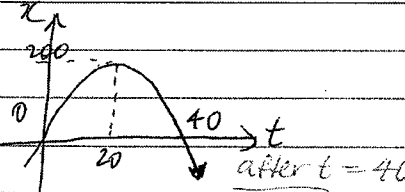
$$0 = -\frac{t^2}{2} + 20t \quad (1)$$

$$0 = t \left(-\frac{t}{2} + 20 \right)$$

$t=0$ initially $t=40$

∴ particle returns to origin after 40 seconds. (1)

iv) graphically



OR after returning to origin particle moves to the left

since at $t=40$ $v = -t + 20 = -20 < 0$ then after $t > 40$ ∴ $v < 0$ and $\ddot{x} = -1$ } keeps moving to the left speeding up.

OR when $t > 40$

$$x = 200 - \frac{(20-t)^2}{2} < 0$$

∴ moves to left

and since v is never zero ∴ keeps moving to the left at increasing speed

2-correct soln

1-attempts to solve eqn. $x=0$

1-correct ans

$x < 0$ & slope is negative ∴ $v < 0$ and $v \neq 0$ & slope is getting steeper ∴ speeding up

(1)

Marks

b) i) $\ddot{x} = 0$

$$\dot{x} = \int 0 dt = c$$

$$\dot{x} = 10$$

$$x = \int 10 dt = 10t + c$$

$$\therefore x = 10t \quad (1)$$

$t=0 \quad v=10$

since $\theta = 0$

$$\therefore \dot{x} = v \cos 0 = v$$

$t=0 \quad x=0 \therefore c=0$

$$\ddot{y} = -g$$

$$\dot{y} = \int -g dt = -gt + c$$

$$\dot{y} = -gt = -10t \quad (1)$$

$t=0 \quad \dot{y} = v \sin 0 = 0$

$\therefore c=0$

$g=10$

$$y = \int -10t dt$$

$$y = -5t^2 + c$$

$t=0 \quad y=200=c$

$$y = -5t^2 + 200 \quad (1)$$

ii) hits the ground ∴ $y=0$

$$\therefore 0 = -5t^2 + 200 \quad (1)$$

$$t^2 = 40 \times t > 0 \therefore t = \sqrt{40}$$

$$\therefore t = 2\sqrt{10} \quad (1)$$

2-correct solns.

1-attempts to solve eqn. for $y=0$

iii) at $t = 2\sqrt{10}$ $\dot{x} = 10$

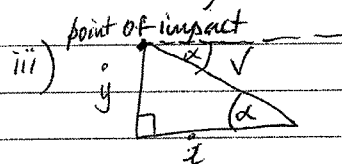
$$\dot{y} = -10t =$$

$$= -10 \times 2\sqrt{10}$$

$$\dot{y} = -20\sqrt{10} \quad (1)$$

You may ask for extra writing paper if you need more space to answer question 12

Question 2b) cont.



$$\dot{x} = 10$$

$$\dot{y} = -20\sqrt{10}$$

3 - correct solns.

$$\text{speed} = |v| = \sqrt{\dot{y}^2 + \dot{x}^2}$$

1 - relates $\dot{x}$ & $\dot{y}$ correctly

$$\text{speed} = \sqrt{(-20\sqrt{10})^2 + 10^2} = \sqrt{4100} = 10\sqrt{41}$$

1 - finds speed

$$\therefore \text{speed} \doteq 64.03 \text{ m/s} \quad \text{OR} \quad 10\sqrt{41}$$

1 - finds acute angle

$$\tan \alpha = \left| \frac{\dot{y}}{\dot{x}} \right| = \left| \frac{-20\sqrt{10}}{10} \right| \quad \alpha - \text{acute}$$

$$\alpha = \tan^{-1} 2\sqrt{10}$$

$$\alpha = 81^\circ 1' \quad 1$$

Question 3

$$a) \ddot{x} = \frac{-1}{4(x-2)^3}$$

3 - correct solns.

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -\frac{1}{4} (x-2)^{-3}$$

2 - finds correct expression for v^2

$$\frac{1}{2} v^2 = -\frac{1}{4} \int (x-2)^{-3} dx$$

$$\frac{1}{2} v^2 = \frac{-1}{4} \frac{(x-2)^{-2}}{-2} + C$$

1 - applies formula $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ correctly

$$\frac{1}{2} v^2 = \frac{1}{8} (x-2)^{-2} + C \quad ①$$

$$x=+3 \quad v=0.5$$

1 - correctly integrates & finds constant

$$+\frac{1}{2}(0.5)^2 = \frac{1}{8} (3-2)^{-2} + C \quad \therefore C=0$$

$$\therefore \frac{1}{2} v^2 = \frac{1}{8} (x-2)^{-2}$$

1 - correctly decides about the sign for velocity

$$\therefore v^2 = \frac{1}{4(x-2)^2} \quad ①$$

$$\therefore v = \pm \frac{1}{2(x-2)}$$

$$\text{but } t=0 \quad x=3 \quad v=+0.5$$

$$\therefore v = + \frac{1}{2(x-2)} \quad ①$$

OR

$$t=0 \quad x=3 \quad v = \frac{1}{2(x-2)} \neq 0$$

$$v=+0.5$$

keeps moving

$$\ddot{x} = -1/4 < 0$$

to left

and for $x > 3$ $v > 0$ & $\ddot{x} < 0$

slowing down $\therefore v \rightarrow 0$

You may ask for extra writing paper if you need more space to answer question 3

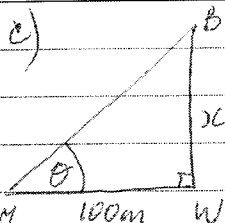
You may ask for extra writing paper if you need more space to answer question 11

Marks

b) i) $x = 1 + 4\cos(3t) + 4\sin(3t)$ $\dot{x} = -4 \times 3\sin(3t) + 12\cos(3t)$ $\ddot{x} = -36\cos(3t) - 36\sin(3t)$ $\therefore \ddot{x} = -9(4\cos(3t) + 4\sin(3t))$ ① where $4\cos(3t) + 4\sin(3t) = x-1$ $\therefore \ddot{x} = -9(x-1)$ ① where $n=3, x_0=1$	2-correct solus. 1-differentiates x & $\dot{x}$ correctly
ii) $T = \frac{2\pi}{n} = \frac{2\pi}{3}$ ① $x = 1 + 4[\cos(3t) + 4\sin(3t)]$ $x = 1 + 4[\sqrt{2}\cos(3t - \alpha)]$ $\alpha = \frac{\pi}{4}$ $\therefore$ amplitude $A = 4\sqrt{2}$ ①	1-correct period 1-correct amplitude
iii) $d_{\text{MAX}} = x_{\text{MAX}} = 1 + 4\sqrt{2}$ ①	1-correct answer
iv) $\ddot{x}$ is max. at extremities $\therefore$ at $x = 1 + 4\sqrt{2}$ or $x = 1 - 4\sqrt{2}$ $\ddot{x} = -9(1 + 4\sqrt{2} - 1) = -36\sqrt{2}$ or $\ddot{x} = -9(1 - 4\sqrt{2} - 1) = +36\sqrt{2}$ $\therefore \ddot{x}_{\text{MAX}} = 36\sqrt{2}$ ①	1-correct answer (accept $\pm 36\sqrt{2}$)

Marks

v) $\ddot{x} = \frac{d}{dt} \left(\frac{1}{2}v^2 \right) = -9(x-1)$ $\frac{1}{2}v^2 = -9 \int (x-1) dx$ $\frac{1}{2}v^2 = -9 \frac{x^2}{2} + 9x + C$ when $x = 1 + 4\sqrt{2}$ $v = 0$ $\therefore v^2 = -9x^2 + 18x + 2C$ $\therefore 0 = -9(1 + 4\sqrt{2})^2 + 18(1 + 4\sqrt{2}) + 2C$ $2C = 279$ $\therefore v^2 = -9x^2 + 18x + 279$ when $x = 2$ $\therefore v^2 = -9(2)^2 + 18(2) + 279$ $\therefore v^2 = 279$ $\therefore \text{speed} = v = \sqrt{279} = 3\sqrt{31}$ or speed = 16.703 cm/s	2-correct answer 1-finds expression for v^2 1-finds equivalent time when $x=2$
(OR) when $x=2$ $2 = 1 + 4\cos(3t) + 4\sin(3t)$ $2 = 1 + 4\sqrt{2}\cos(3t - \frac{\pi}{4})$ $\frac{1}{4\sqrt{2}} = \cos(3t - \frac{\pi}{4})$ $36 - \frac{\pi}{4} = 1.393 \dots$ (sec) $t = 0.72616 \dots$ $\therefore \dot{x} = -12(\sin 3t - \cos 3t) $ $\therefore \text{speed} = 16.703 \text{ cm/s}$	

c)  $V = \frac{dx}{dt} = 5 \text{ m/s}$	3 - correct solus
i) $\tan \theta = \frac{x}{100} \therefore x = 100 \tan \theta$ $\frac{dx}{d\theta} = 100 \sec^2 \theta$	2 - finds correct expression for $\frac{d\theta}{dx}$ or $\frac{dx}{d\theta}$ and attempts to find $\frac{d\theta}{dt}$
OR $\tan \theta = \frac{x}{100}$ $\theta = \tan^{-1} \frac{x}{100}$ $\frac{d\theta}{dx} = \frac{1}{100} \times \frac{1}{1 + \left(\frac{x}{100}\right)^2}$ OR $\frac{100}{100^2 + x^2}$	1 - finds another rate relevant to the problem
$\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$ $= \frac{1}{100} \times \frac{1}{1 + \left(\frac{x}{100}\right)^2} \times 5$ ① $= \frac{5}{100 \times \frac{100^2 + x^2}{100^2}} = \frac{500}{100^2 + x^2}$ ①	
ii) when $\theta = \frac{\pi}{4} \therefore x = 100 \times \tan \frac{\pi}{4}$ $\therefore x = 100$ $\therefore \frac{d\theta}{dt} = \frac{500}{100^2 + 100^2} = \frac{1}{40} \text{ rad/sec.}$	1 - correct answer with correct units

You may ask for extra writing paper if you need more space to answer question 3